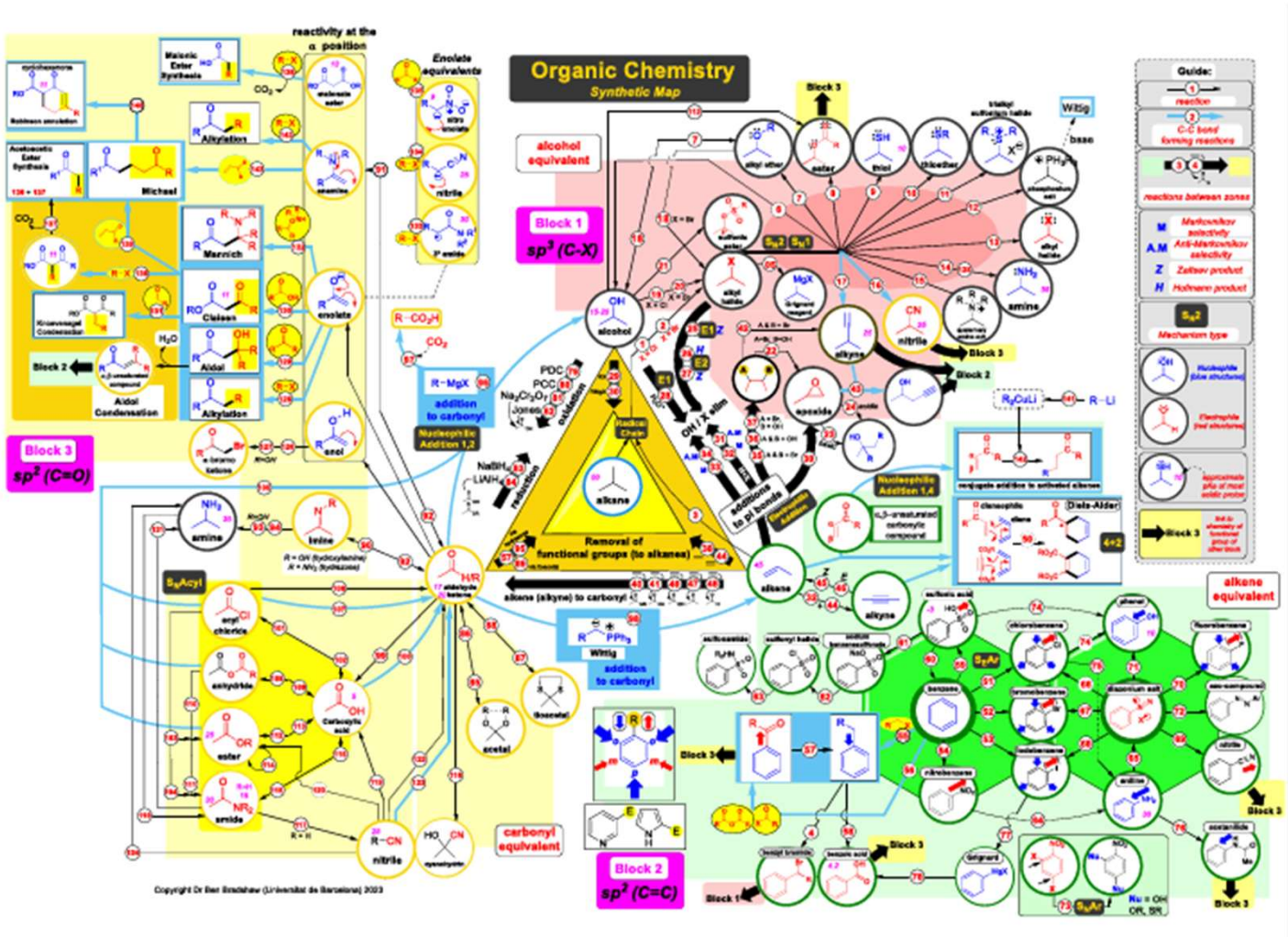
https://www.syntheticmap.com/



“Synthetic Map” showing the relationship and connectivity between different functional groups. Copyright Dr. Ben Bradshaw (Universitat de Barcelona)

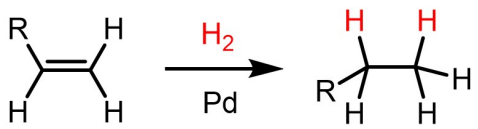
Tentative Schedule (1) Revised

Week	Date (Mon)	Lecture (Wed)	Lecture (Fri)	Tutorial
1	10/8	Electrophilic addition of alkenes	Electrophilic addition of alkenes	
2	17/8	Electrophilic aromatic substitution	Electrophilic aromatic substitution	
3	24/8	Reduction	Oxidation/Alkene Synthesis	Tutorial 1 (Electrophilic addition of alkenes; Electrophilic aromatic substitution)
4	31/8	Enolate Chemistry (alkylation reaction)	Enolate Chemistry (Aldol reactions)	Tutorial 2 (Electrophilic aromatic substitution; Oxidation/reduction/Alkene Synthesis)
5	7/9	Enolate Chemistry (Aldol reactions; Claisen)	Enolate Chemistry (Enol and enamine) Practice	Tutorial3 (Enolate Chemistry-alkylation reaction/Aldol reactions)
6	14/9	Conjugate addition	**Test 1** (covers weeks 1 to 5)	

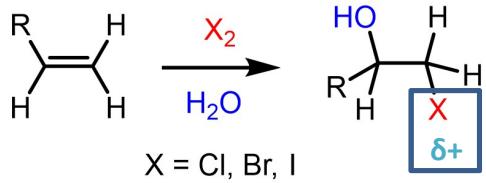
Tentative Schedule (2) Revised

Week	Date (Mon)	Lecture (Wed)	Lecture (Fri)	Tutorial
7	28/9	Review Test 1; Protecting groups	Cycloadditions	Tutorial 4 (Enol and enamine, conjugated addition) **Fri**
8	5/10	Cycloadditions	NUS wellbeing day	Tutorial 4 (Enol and enamine, conjugated addition) **Tue**
9	12/10	Cycloadditions	Sigmatropic rearrangements	Tutorial 5 (cycloaddition) **Fri**
10	19/10	Electrocyclic reaction	Radical chemistry	Tutorial 5 (cycloaddition) **Tue** Tutorial 6 (sigmatropic; electrocyclic) **Fri**
11	26/10	Radical chemistry	**Test 2** (covers week 6 to week 10, excluding radical chemistry)	Tutorial 6 (sigmatropic; electrocyclic) **Tue**
12	2/11	Radical chemistry/Cross-coupling reactions	Cross-coupling reactions	
13	9/11 Deepavali	Review Test 2; Practice (Radicals, Cross-coupling etc.)	(consultation)	

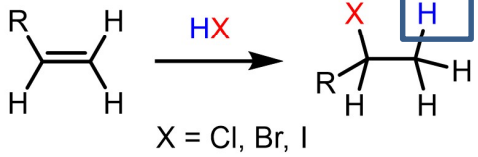
Electrophilic Addition Reactions



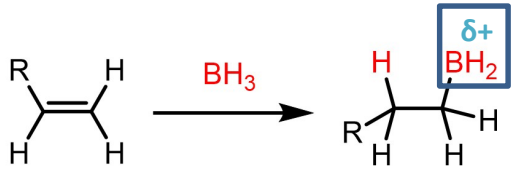
Hydrogenation



Halohydrin formation

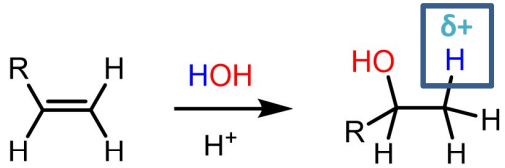


Hydrohalogenation

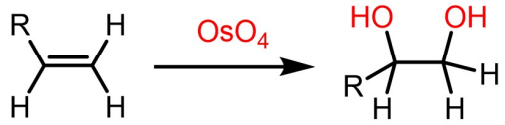


Hydroboration

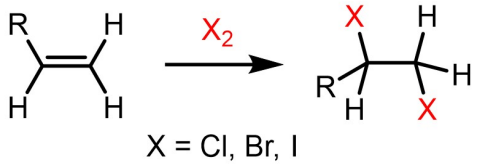
continues



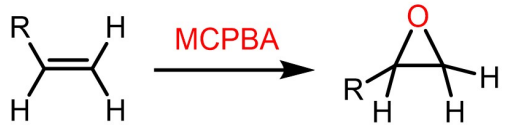
Hydration



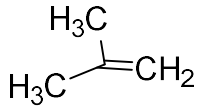
Dihydroxylation



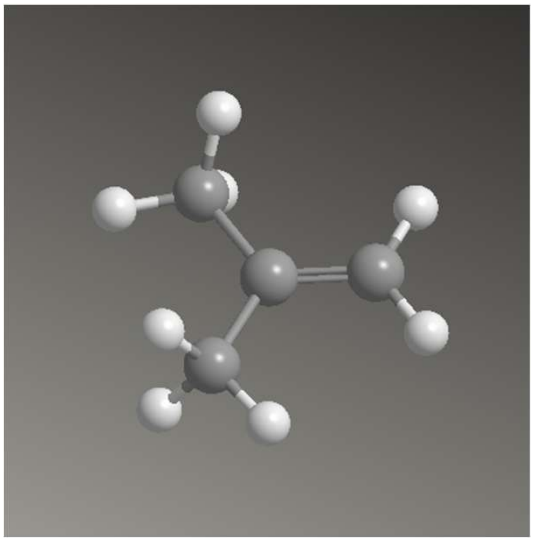
Halogenation

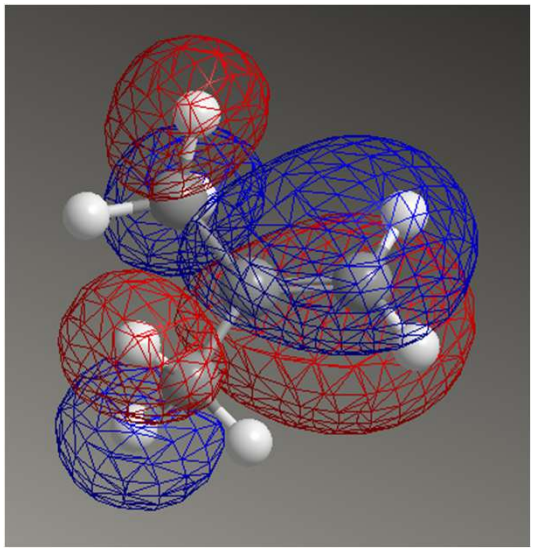


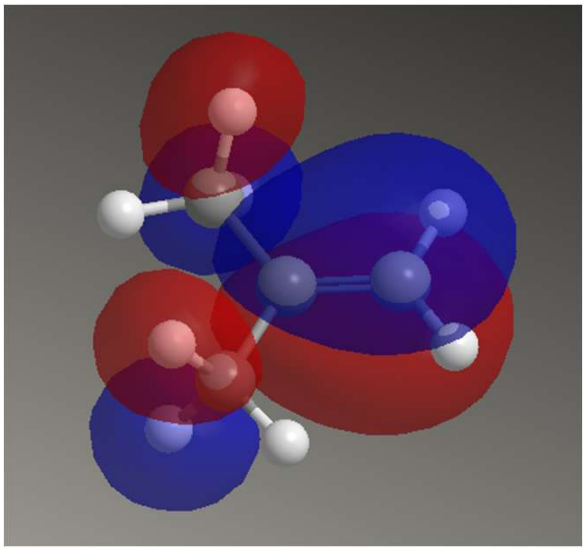
Epoxidation

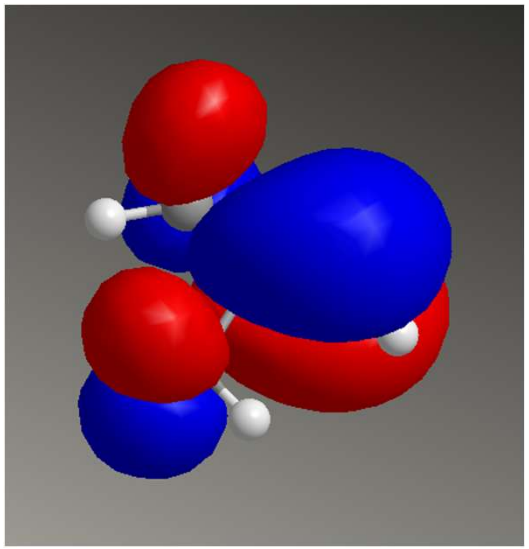


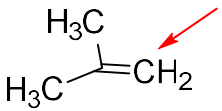
HOMO of isobutene











high electron density

CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

Office: S9-14/Tel: 66017501

chmzhu@nus.edu.sg

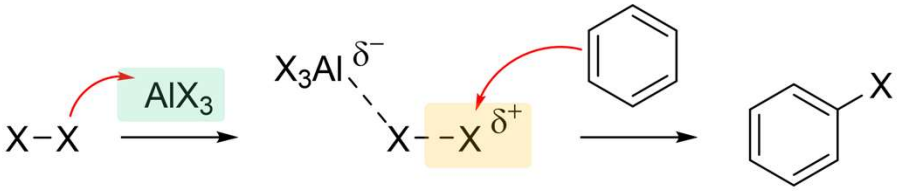
Electrophilic Aromatic Substitution (Prat 1)

The aims of this chapter are …..

- electrophilic attack on benzene
- activation and deactivation of the benzene ring
- position of substitution on the benzene ring

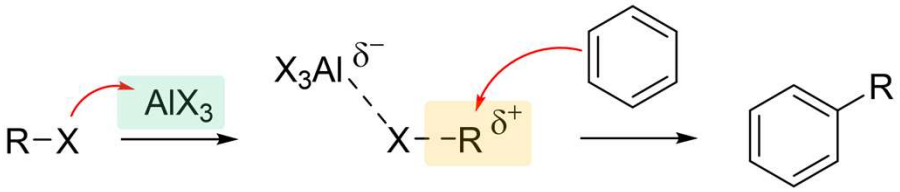
Electrophilic Aromatic Substitution

1. Bromination or **chlorination**

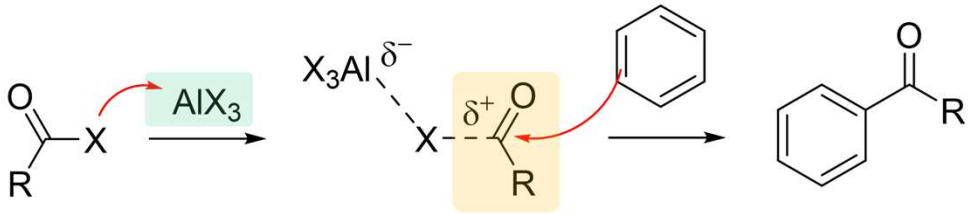


X = Br or Cl; AlX_3 can be other **Lewis acid** (e.g., FeX_3)

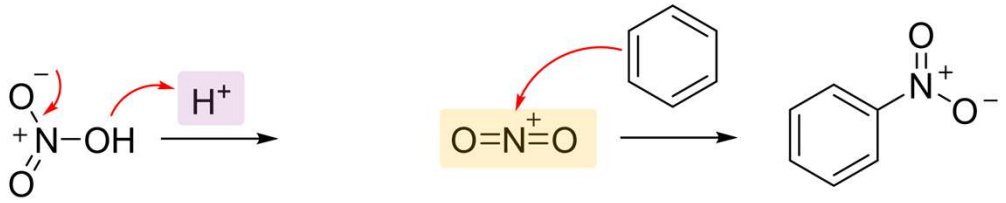
2A. Friedel-Crafts alkylation



2B. Friedel-Crafts acylation

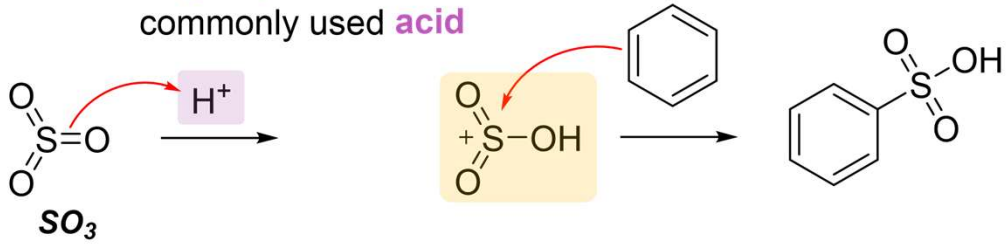


3. Nitration



HNO_3 **H_2SO_4** is the most commonly used **acid**

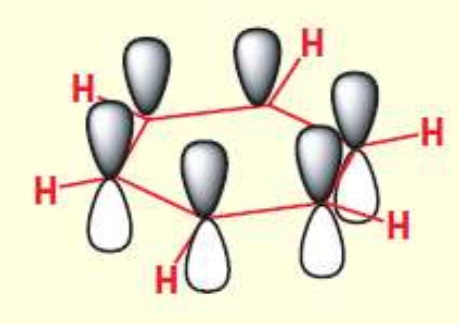
4. Sulfonation



SO_3

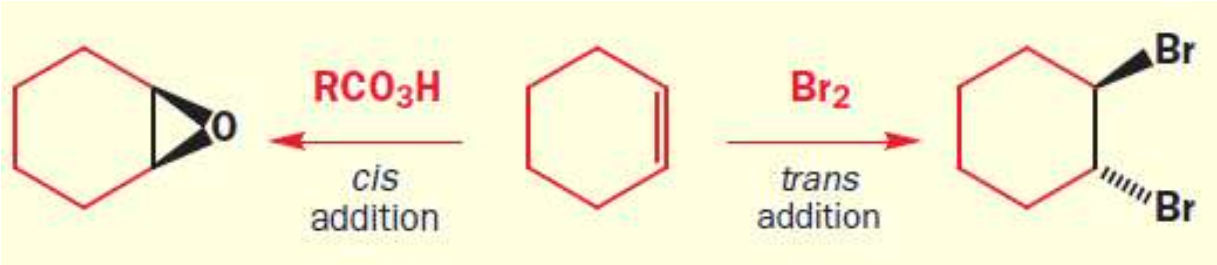
1. Benzene and Its Reaction with Electrophiles

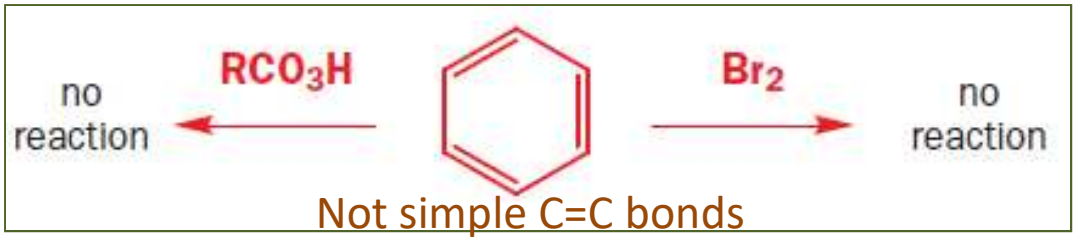
Benzene is a planar symmetrical hexagon with six trigonal (sp^2) carbon atoms, each having one hydrogen atom in the plane of the ring. All the bond lengths are 1.39 Å (compare C–C 1.47 Å and C=C 1.33 Å).



The special stability of benzene (aromaticity) comes from the six π electrons in three molecular orbitals made up by the overlap of the six atomic p orbitals on the carbon atoms.

Recall: Simple alkenes react rapidly with electrophiles such as peroxyacids or bromine

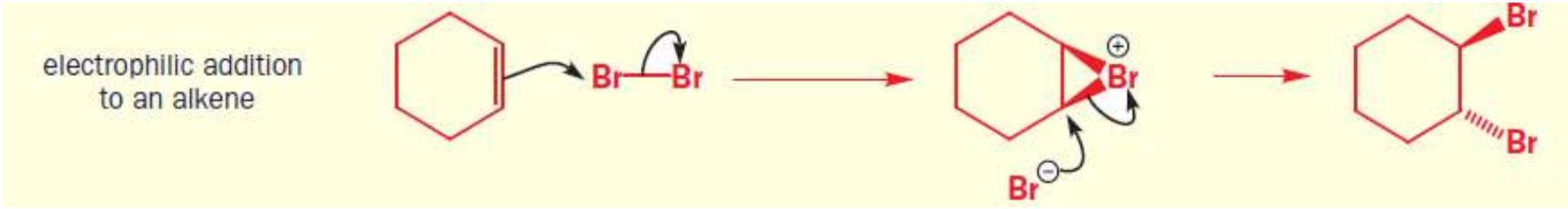


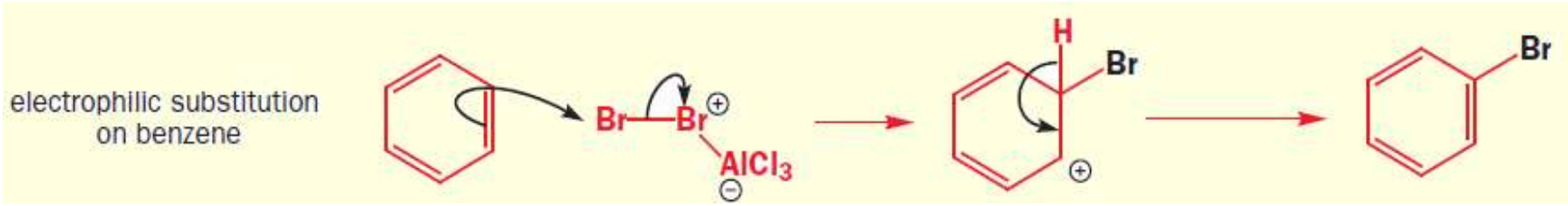


Not simple C=C bonds

Halogenation reactions of Benzene vs Cyclohexene

Benzene can react with bromine if a Lewis acid catalyst such as **$AlCl_3$** and **$FeCl_3$** is added.

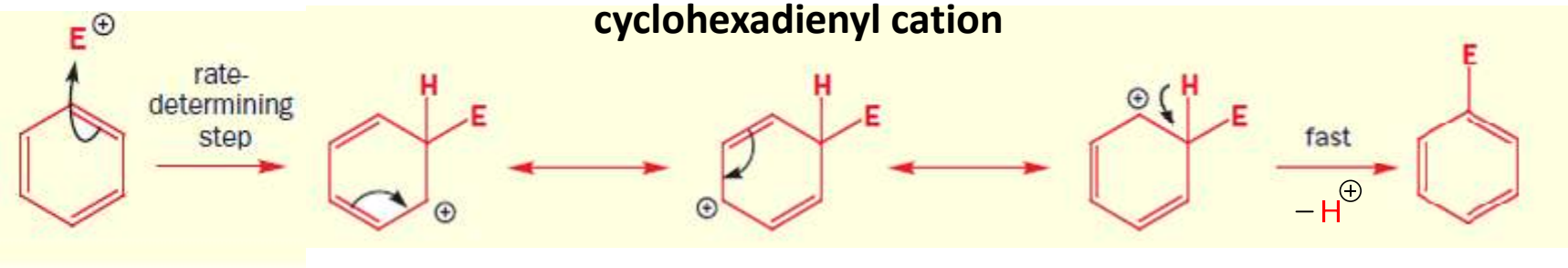




Bromine atom has replaced an atom of hydrogen and so this is a substitution reaction. The reagent is electrophilic bromine and the molecule is aromatic so the reaction is **electrophilic aromatic substitution**

2. Mechanism for Electrophilic Aromatic Substitution

The charge can be delocalized to the two *ortho positions and the para position*



Take note of where the (+) could be located

The cation can be represented as three different delocalized structures that show clearly the electron-deficient atom



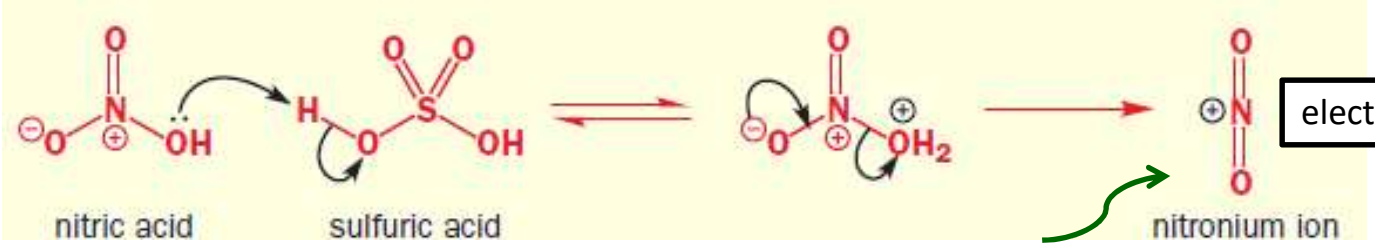
terminology

3. Nitration of Benzene

Introduction of a nitro (NO_2) group into an aromatic system provides a general entry into aromatic nitrogen compounds (this reaction is not available for aliphatic nitrogen compounds).

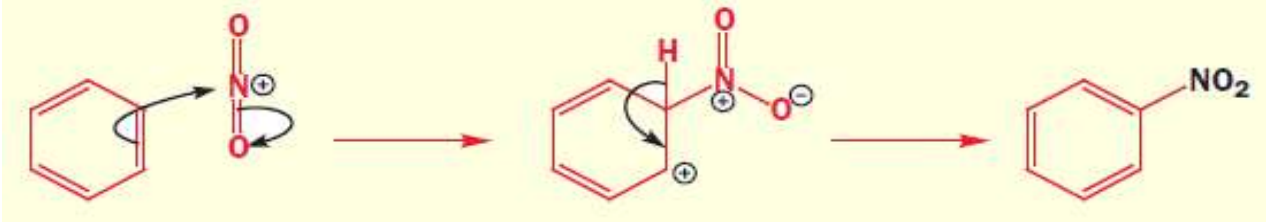
Aromatic nitration requires very powerful reagents, typically a mixture of conc. nitric and sulfuric acids.

1st step: formation of a very powerful electrophile NO_2^+, by the interaction two strong acids



linear with an sp hybridized nitrogen at the centre

Benzene attacks the positively charged nitrogen atom and one of the N=O bonds must be broken at the same time to avoid five-valent nitrogen



Exploiting the chemistry of the nitro group

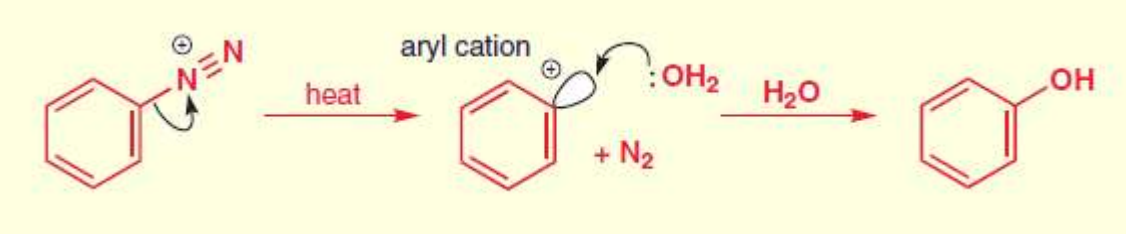
- can be reduced to an amino group;



- The resulting anilines can undergo diazonium chemistry



Mechanism



N_2 is a good leaving group. So upon warming, it departs leaving behind a cation which is captured by a nucleophile (in this case – water) to provide phenol

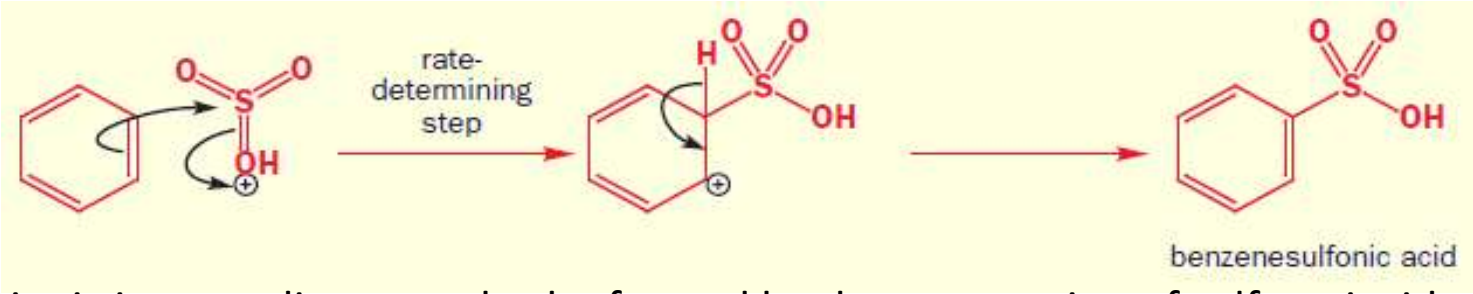
4. Sulfonation of Benzene

Benzene reacts slowly with sulfuric acid alone to give benzenesulfonic acid



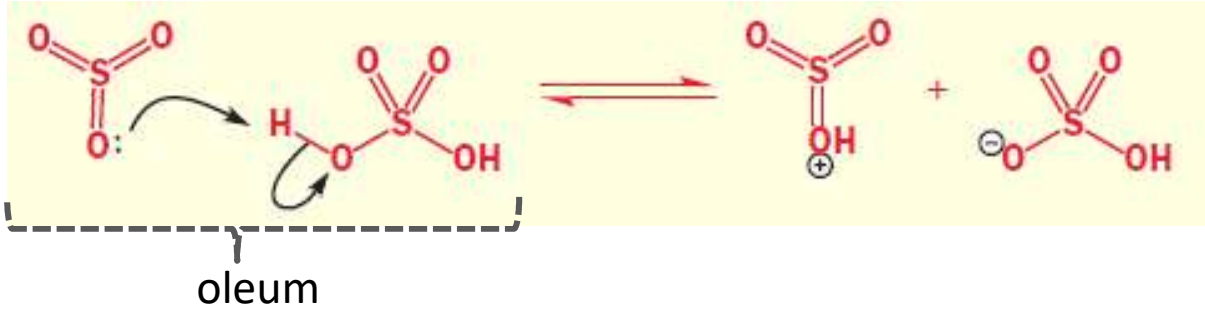
electrophile

The cation produced is very reactive and combines with benzene by the same mechanisms we have seen for bromination and nitration



The cationic intermediate can also be formed by the protonation of sulfur trioxide, SO_3.

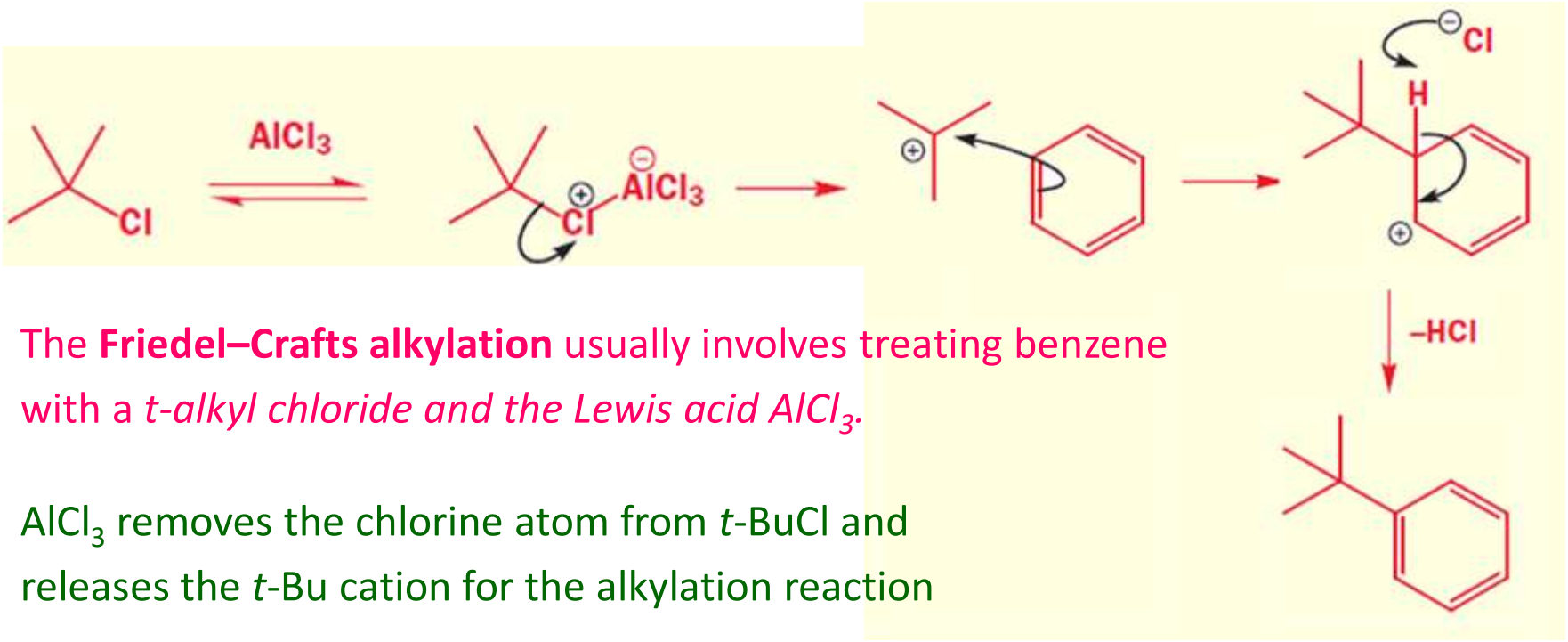
So another way to do sulfonations is to use concentrated sulfuric acid with SO_3



5. Friedel–Crafts Alkylation

Recall: any nucleophile, however weak, will react with a carbocation in the S_N1 reaction and even benzene rings will do this

Adding C electrophiles requires reactive C electrophiles ➡ carbocations



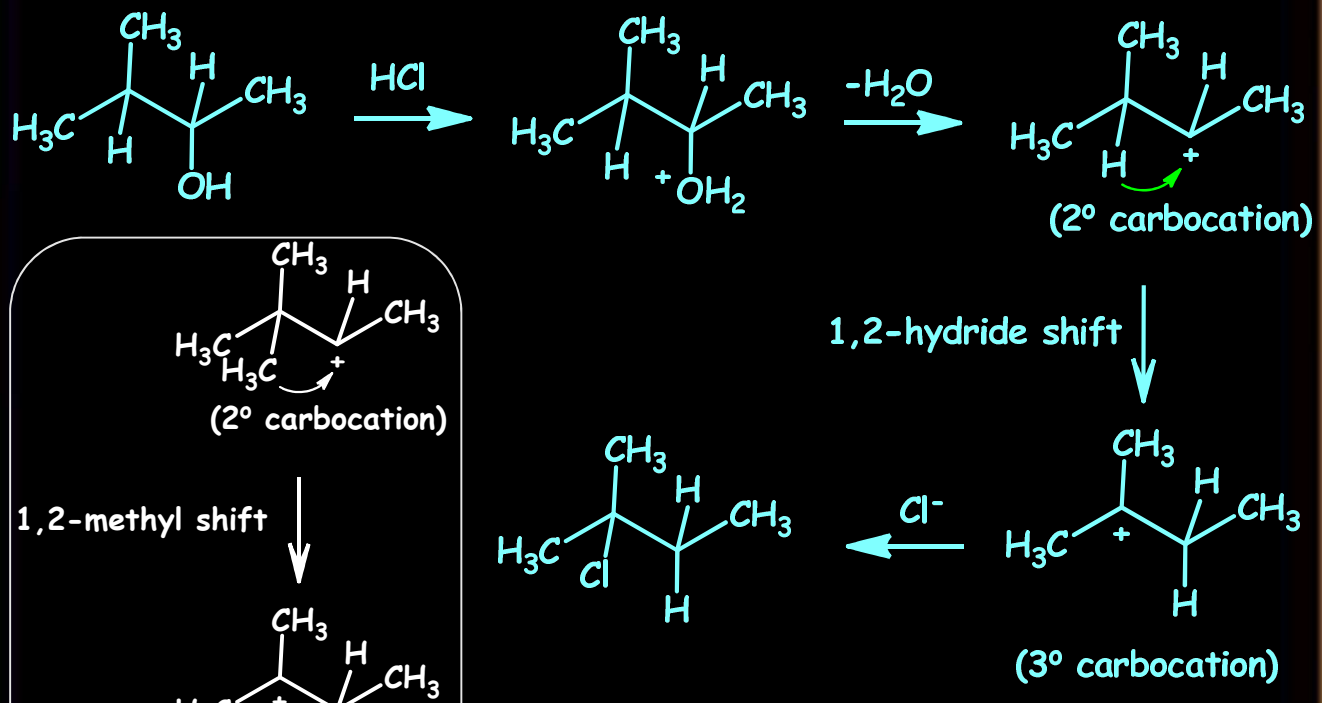
The **Friedel–Crafts alkylation** usually involves treating benzene with a *t-alkyl chloride and the Lewis acid* $AlCl_3$.

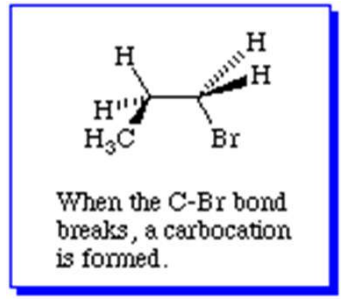
$AlCl_3$ removes the chlorine atom from *t*-BuCl and releases the *t*-Bu cation for the alkylation reaction

In the alkylation step it is essential that the alkyl group can form a cation; otherwise the reaction does not work very well

Carbocations can readily rearrange

Carbocation stability: $CH_3^+ < CH_3CH_2^+ < (CH_3)_2CH^+ < (CH_3)_3C^+$

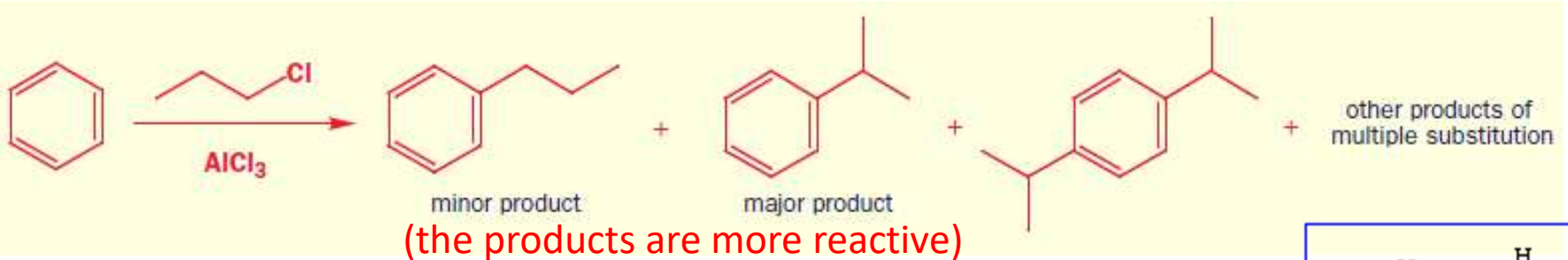




1,2-hydride shift

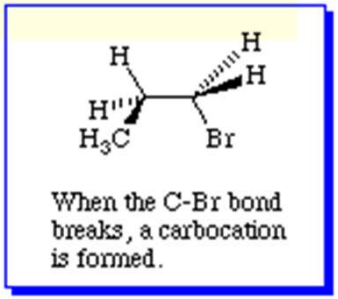
Friedel–Crafts alkylation can provide the rearranged product

alkyl cations often rearrange to yield more stable cations

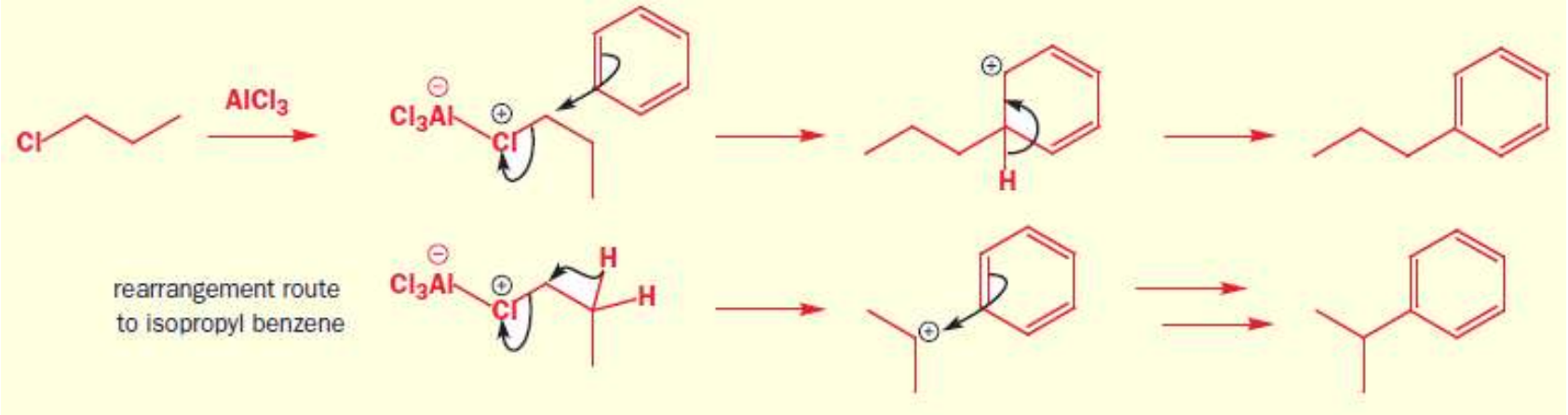


(the products are more reactive)

The rearrangement in this mechanism occurs because 1° cations do not exist in solution so that the alkyl halide–$AlCl_3$ complex must either react directly or rearrange to the more stable 2° carbocation. Similarly if a 2° cation can rearrange to a 3° carbocation, the rearrangement will occur.



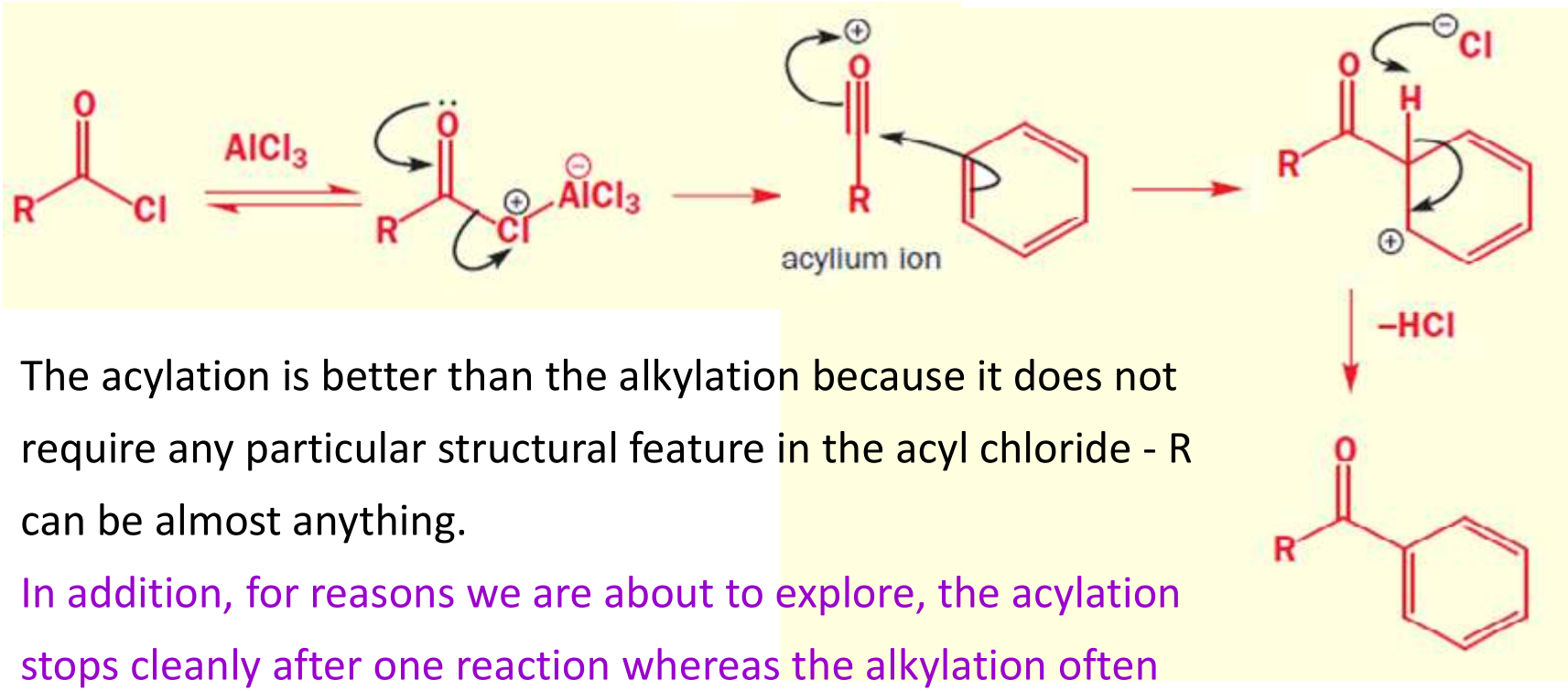
1,2-hydride shift



6. Friedel–Crafts Acylation

Friedel–Crafts acylation usually involves treating benzene with acid *chloride and the Lewis acid* $AlCl_3$.

Acid chlorides can give the rather stable acylium ions

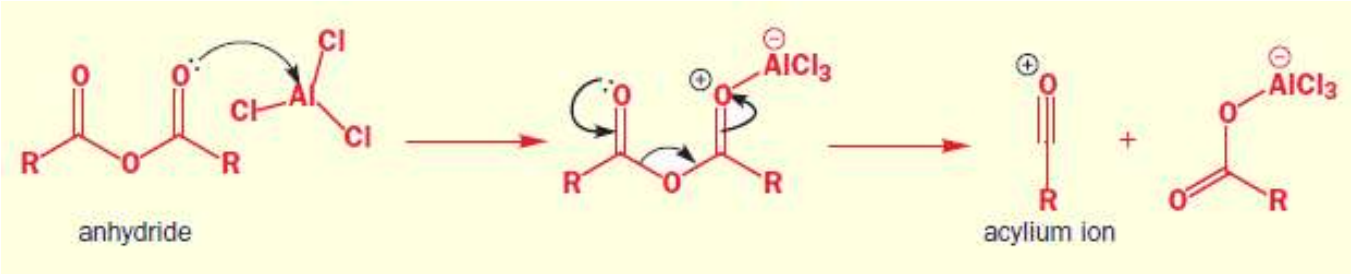


The acylation is better than the alkylation because it does not require any particular structural feature in the acyl chloride - R can be almost anything.

In addition, for reasons we are about to explore, the acylation stops cleanly after one reaction whereas the alkylation often gives mixtures of products.

Friedel–Crafts *acylation*

Besides acyl chloride, an acid anhydride is used together with the Lewis acid to produce the reactive acylium ion.



Multiple substitution is rarely a problem because the deactivated conjugated ketone is much less reactive than benzene

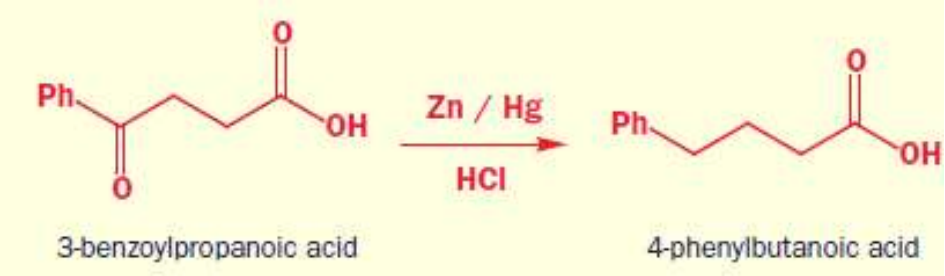


Because the acylation reaction is so much more reliable than Friedel-Crafts alkylation, a common method to alkylate is actually acylation first and then reduce the carbonyl to a methylene group ($–CH_2–$)

- simple reduction with zinc amalgam (an alloy of zinc and mercury) in HCl converts the ketone to an alkylated product (Clemmensen reduction)
- particularly effective at reducing aryl-alkyl ketones



The reduction of ketone is chemo-selective (e.g., acid does not react)

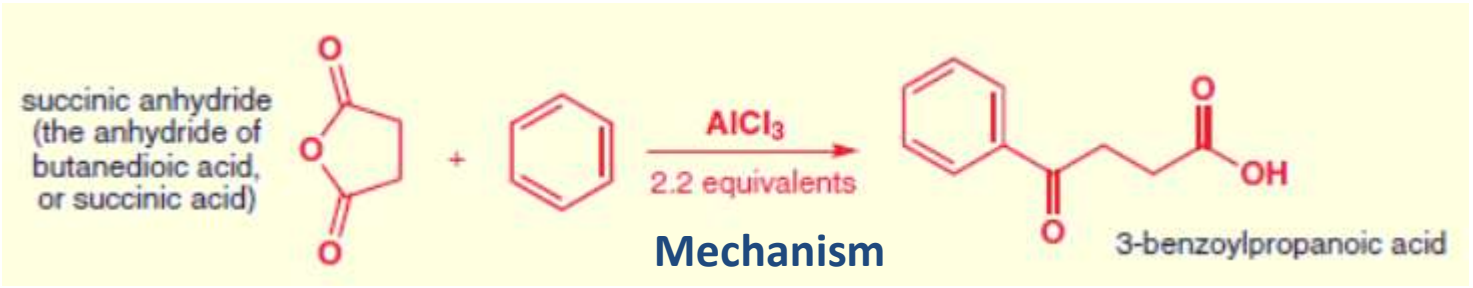


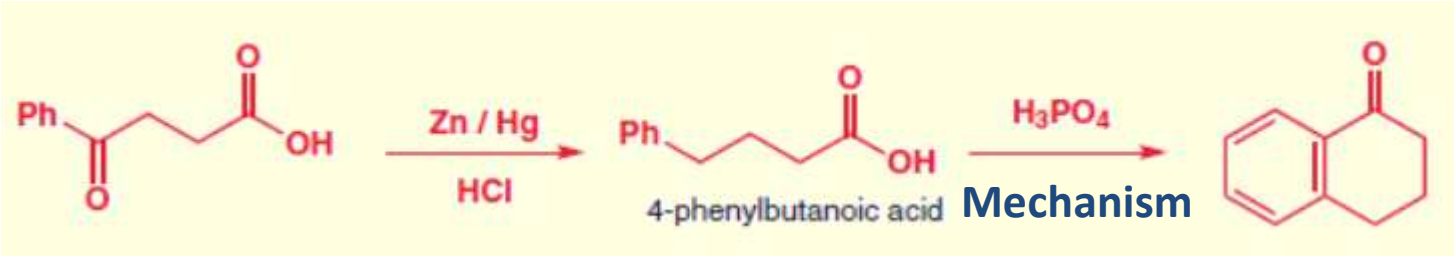
acylation vs alkylation

Two problems in Friedel-Crafts alkylation do not arise with acylation:

• the acyl group in the product withdraws electrons from the π system making multiple substitutions harder. In fact, if the ring is too deactivated to start off with, Friedel-Crafts acylation may not be possible at all *eg* nitrobenzene is inert to Friedel–Crafts acylation and is often used as a solvent for these reactions

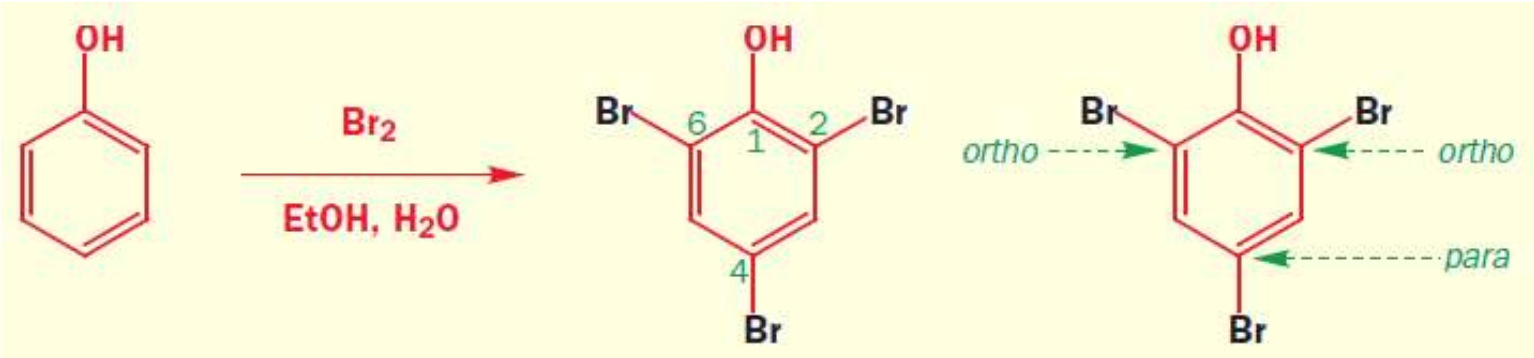
• Rearrangements are also no longer a problem because the electrophile, the acylium cation, is already relatively stable


succinic anhydride
(the anhydride of
butanedioic acid,
or succinic acid)
AlCl3
2.2 equivalents
Mechanism
OH
3-benzoylpropanoic acid


Ph
OH
Zn / Hg
HCl
4-phenylbutanoic acid
H3PO4
Mechanism

7. Electrophilic Substitution on Phenols

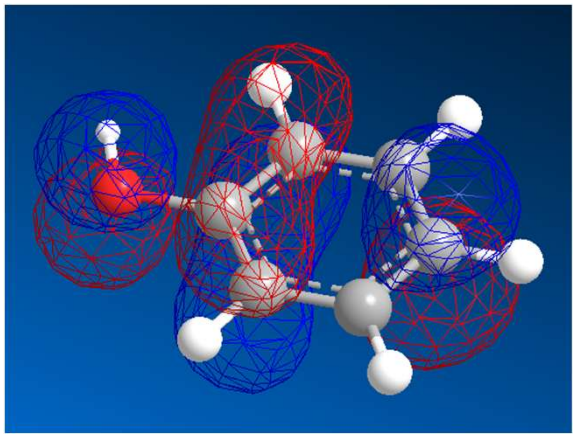
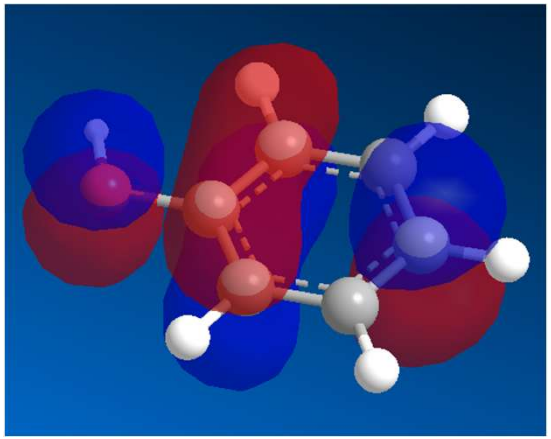
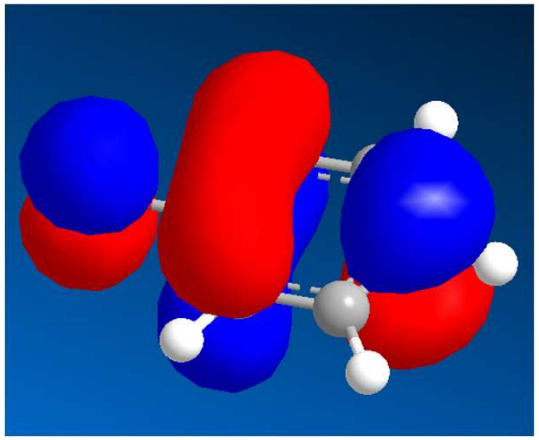
Benzene does not react with bromine except with Lewis acid catalysis. Phenols react in a very different manner: no Lewis acid is needed, the reaction occurs very rapidly, and the product contains three atoms of bromine in specific positions.

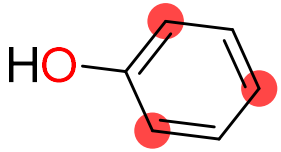


Phenol reacts three times as rapidly without catalyst at room temperature. This is because:

- the highest-energy electrons in phenol are no longer those in the benzene ring but the lone pairs on oxygen.
- these nonbonding electrons contribute to a much higher-energy HOMO than the very low-energy bonding electrons in the aromatic ring

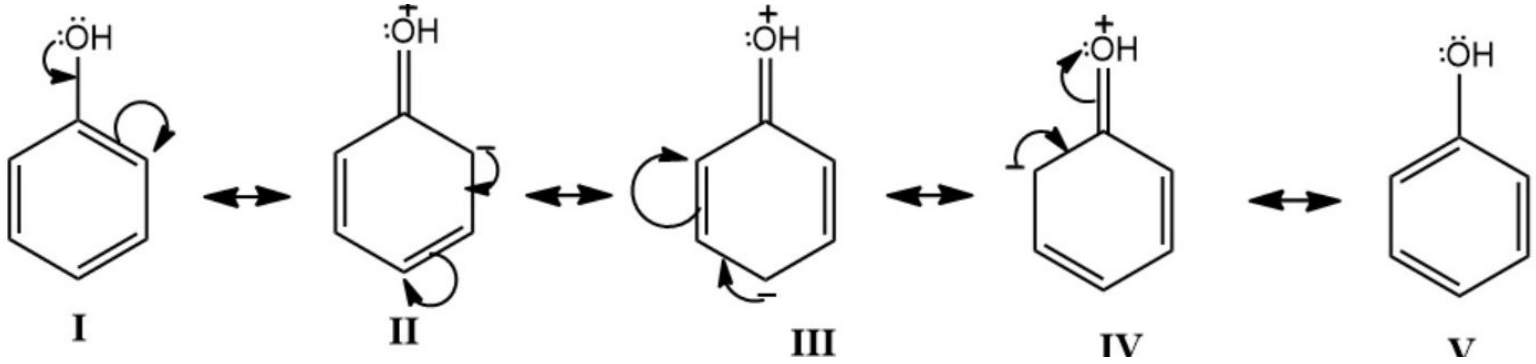
HOMO of Phenol

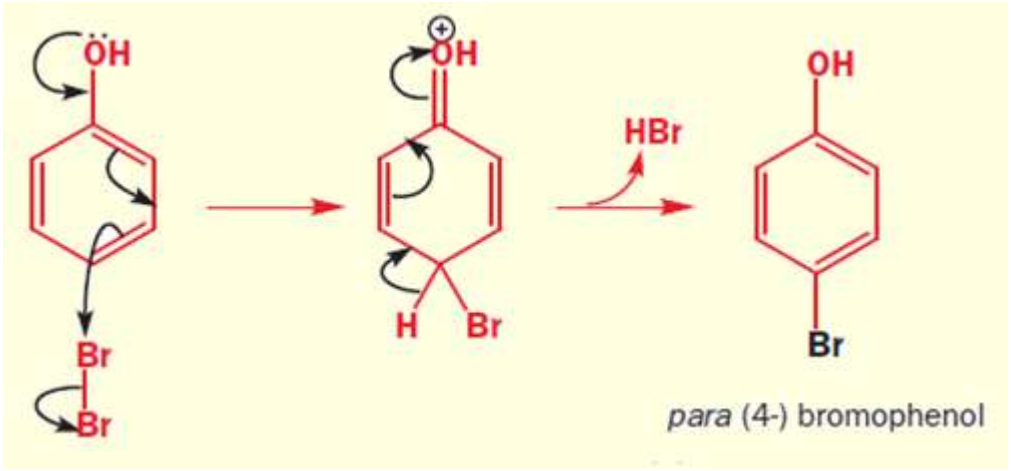






high electron density

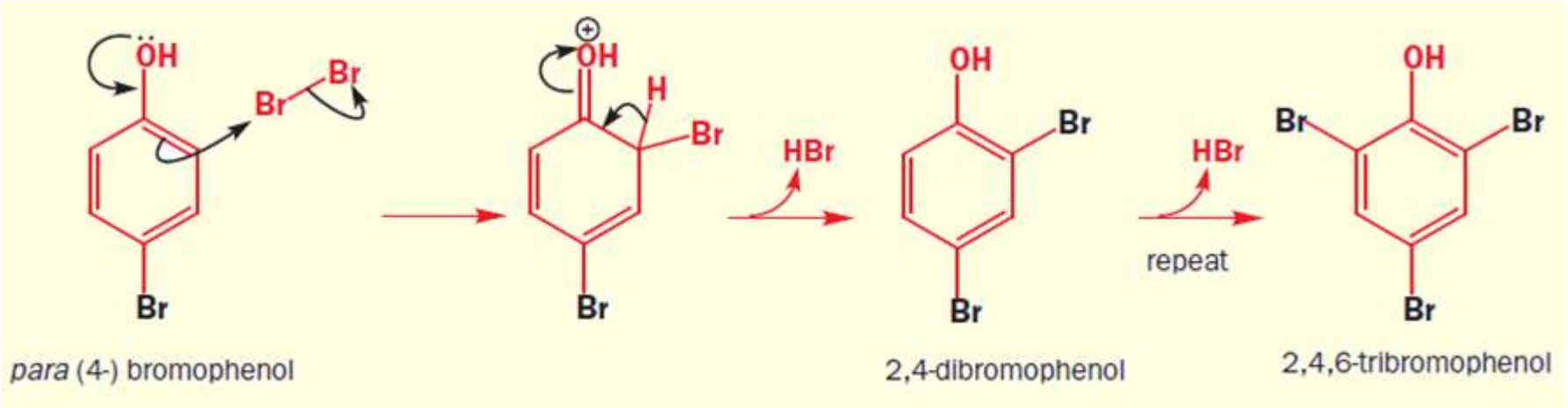
Explanation using resonance structures





start the chain of arrows from the lone pair electrons on the OH and push them through the ring. The benzene ring is acting as a conductor allowing electrons to flow from the OH group to the bromine molecule.

Mono bromination: *para*-product and *ortho*-product

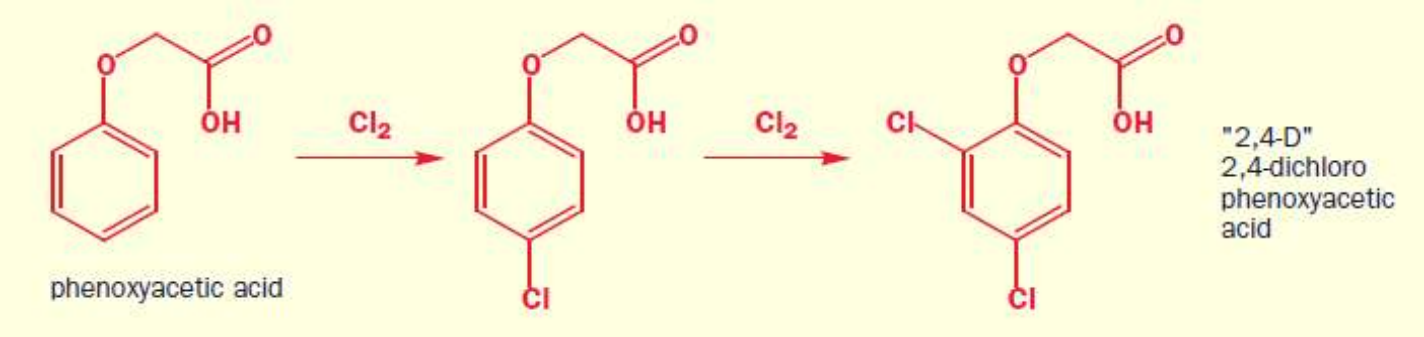


The OH group is said to be ***ortho, para-directing towards electrophiles.***

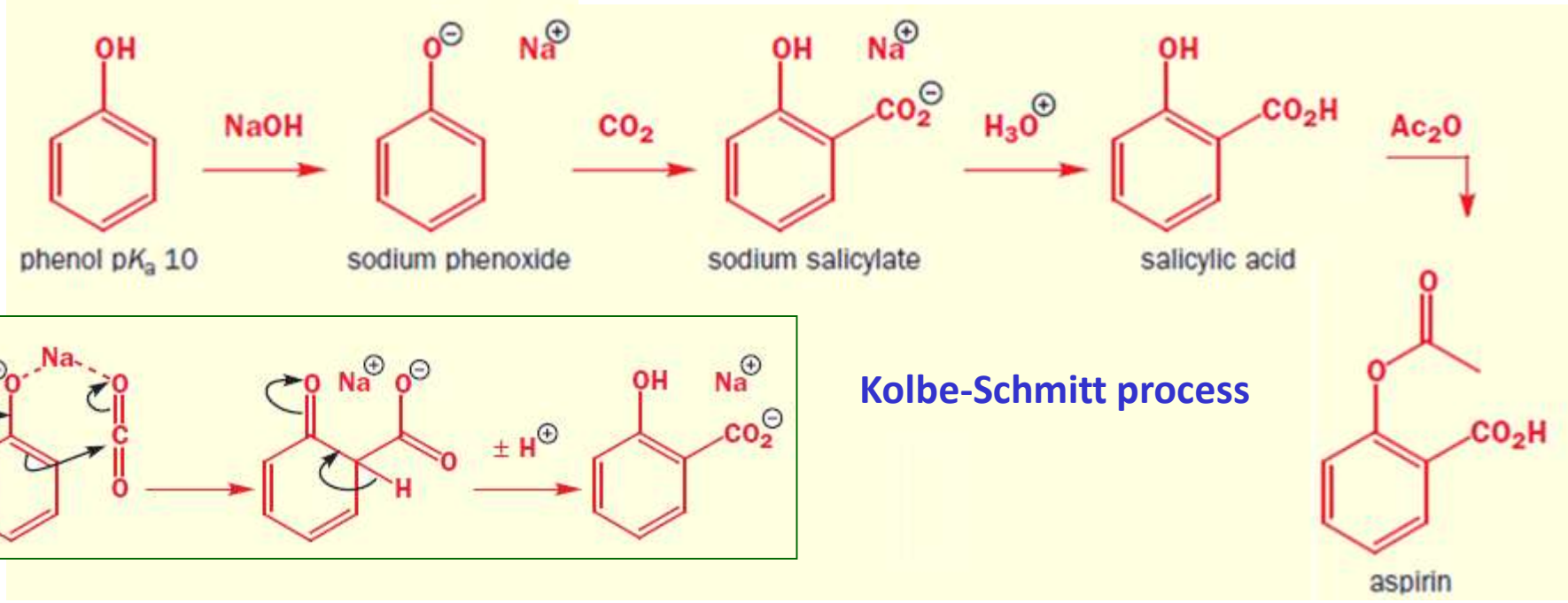
Applications

Ether of phenol

1st chlorination at para due to steric hindrance



The **phenoxide ion** is more reactive (draw the resonance structure) towards electrophilic attack. It will even react with such weak electrophiles as carbon dioxide.



Kolbe-Schmitt process

Summary

- Electrophilic aromatic substitution reactions are a class of transformations that form functionalized arenes.
- Mechanism of the reactions (cyclohexadienyl cation as intermediate)
- Halogenation
- Nitration
- Sulfonation
- Friedel–Crafts alkylation
- Friedel–Crafts acylation
- Electrophilic Substitution on phenols (*para*- and *ortho*-reactions are favored)